

smlCA - single-molecule Image to Concentration Analyser

This repository provides software for calculations of fluorophores' concentrations from the FLIM images stored in the PTU format (PicoQuant - Germany).

About

The Software allows the extraction of the information about registered photons from the *.ptu* file, filtering them using statistical filters based on fluorescence decay time, autodetect region of interest (ROI), and finally, creating the map of the absolute concentration of the fluorophore.

List of changes

1. The two previously separated scripts (*EXTRACT_AND_FILTER_PTU* and *Phot2Conc*) has been integrated into one script. At the startup, the user can choose the operation mode (EXTRACT from PTU and FILTER or Phot 2 Conc). The *REWRITE_ROI* script is still available as the users still have a chance to load the ROI file that was created externally as the text image. Since the external ROI file is not crucial in the analysis, the script is in the *src* folder. After installation, the *REWRITE_ROI* script is located in the *smlCA* folder at the installation path.
2. The code was modified and polished
3. In the Phot 2 Conc mode, the automatic ROI feature was added, automatically detecting cells, removing the nucleus from the cell, or copying the ROI between channels.
4. Minor graphical modifications and adjustments.
5. Installation setup for Linux (tested on Fedora 40) and Windows (tested on Windows 10 and 11).

Installation guide

1. Download the *smlCA_installation_files.zip* file and extract in the desired location.
2. Run the installation script:
 - A. On Linux, run the *install.sh*, shell script by typing:

```
./install.sh
```
 - B. On Windows, double-click the *install_win.bat* executable script.

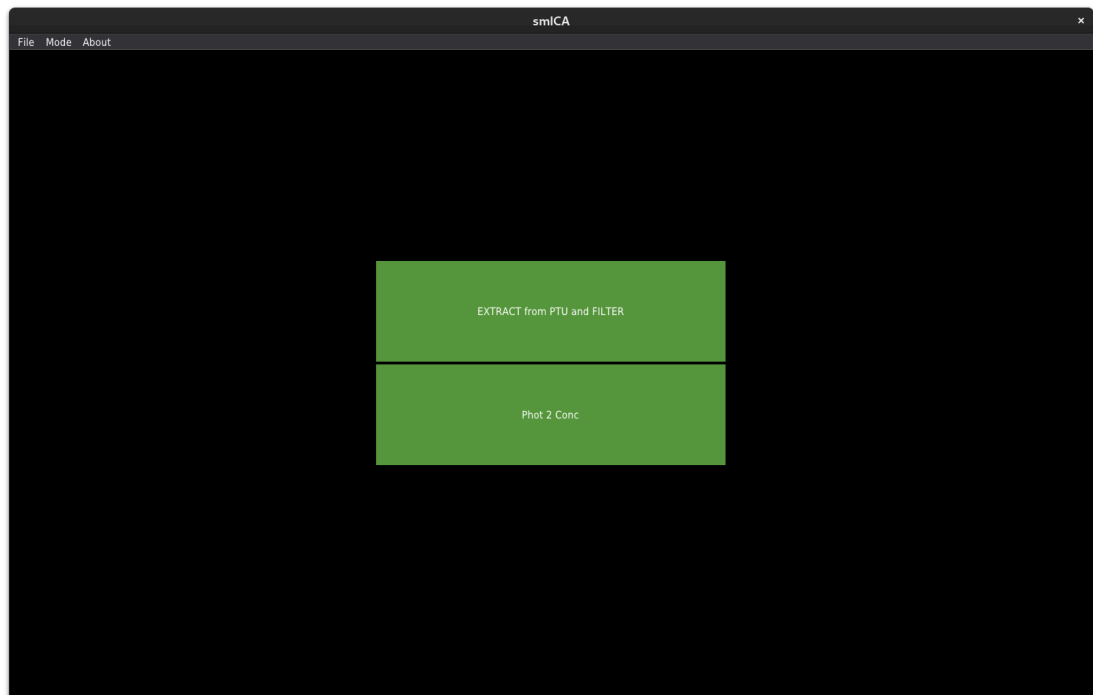
The installation can take a while as all required packages must be downloaded.

User guide

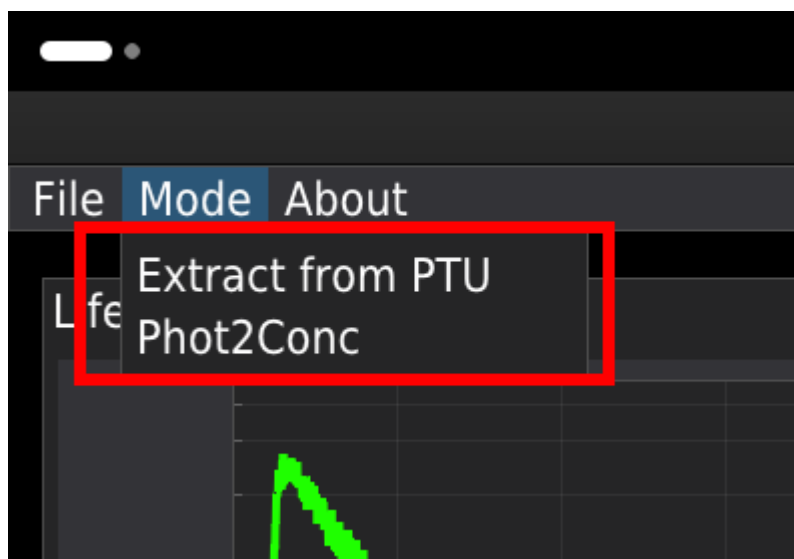
The workflow for mapping the fluorophore's concentration in living cells can be divided into two to three steps, depending on the user preferences.

1. Extract the information about photons from the binary *.ptu* file and filter the data to eliminate background or unwanted influence of autofluorescence or other fluorophores.
2. (OPTIONAL) Create the ROI file using external software, for example, ImageJ, to create a binary text image containing a selection of the pixels that should be included in the analysis. This step is optional because the third step has an option for automatic detection of the cells. A detailed description of using the *REWRITE_ROI* script can be found in the *README* file in the script's folder.
3. Calculate the map of the fluorophore concentration based on fluorescence intensity data and calibration data.

At the startup, the user can choose between two modes: [EXTRACT from PTU and FILTER](#) or [Phot 2 Conc](#).

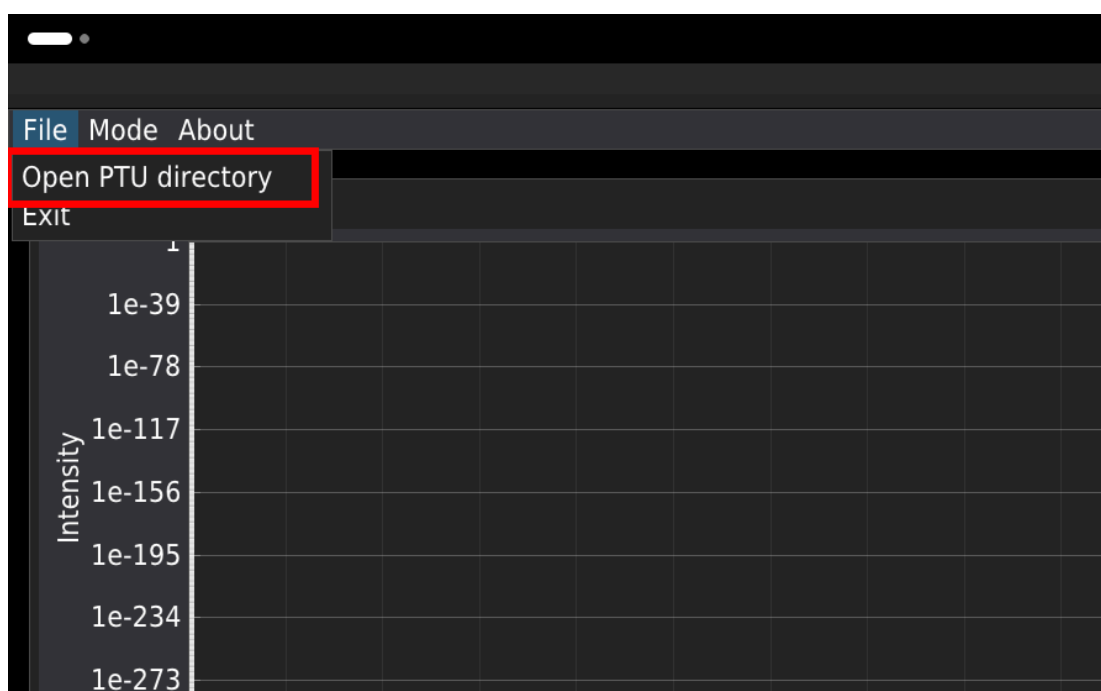


The first option is to extract raw data from the *.ptu* files, filter the data using time gating, or apply statistical filters based on the fluorescence decay pattern. The second option provides a GUI to map the concentration of fluorescent molecules inside living cells. Below is a detailed description of each mode. You can also switch between modes by selecting the proper option from the Mode menu.



EXTRACT from PTU and FILTER

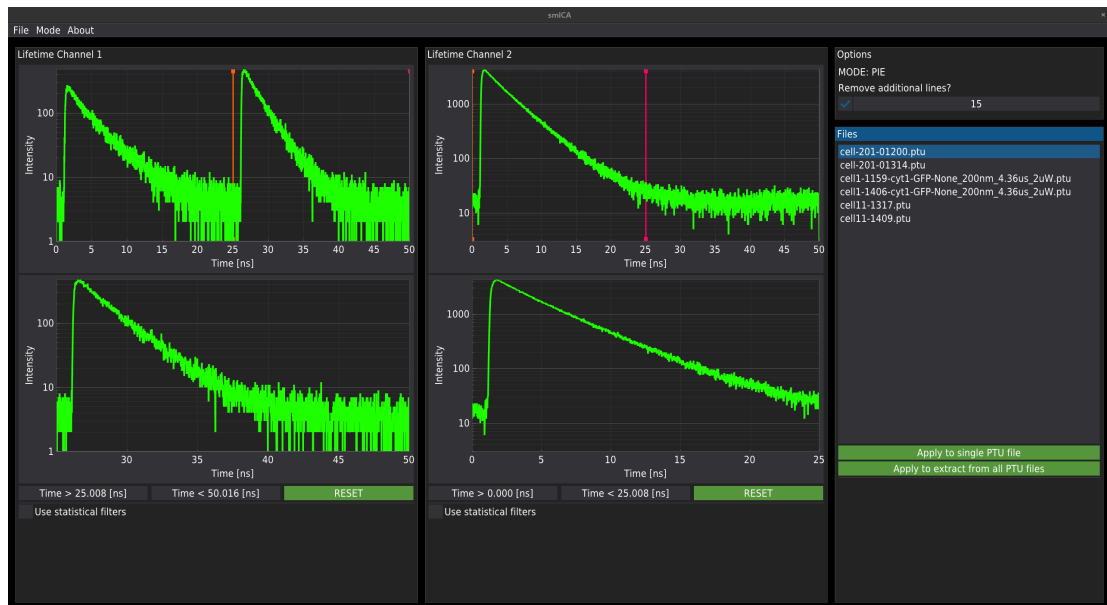
First, click "File" and "Open PTU directory" to select the folder storing your *.ptu* files.



Keeping the *.ptu* from different experiments in separate folders is suggested.

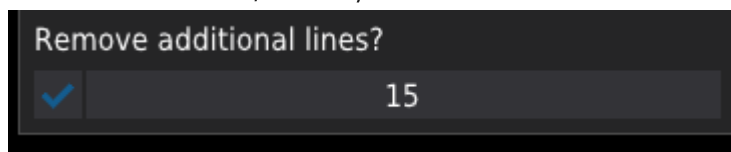
The script recognizes the mode in which the measurement was performed (standard or PIE). If two detectors are used in the measurement, the signal will be automatically split and displayed as Channel 1 and Channel 2. The lifetime analysis can be performed for each channel separately.

Simple time gating can be realized by selecting the region of the fluorescence decay pattern by moving the orange and red drag lines displayed in the upper plots.



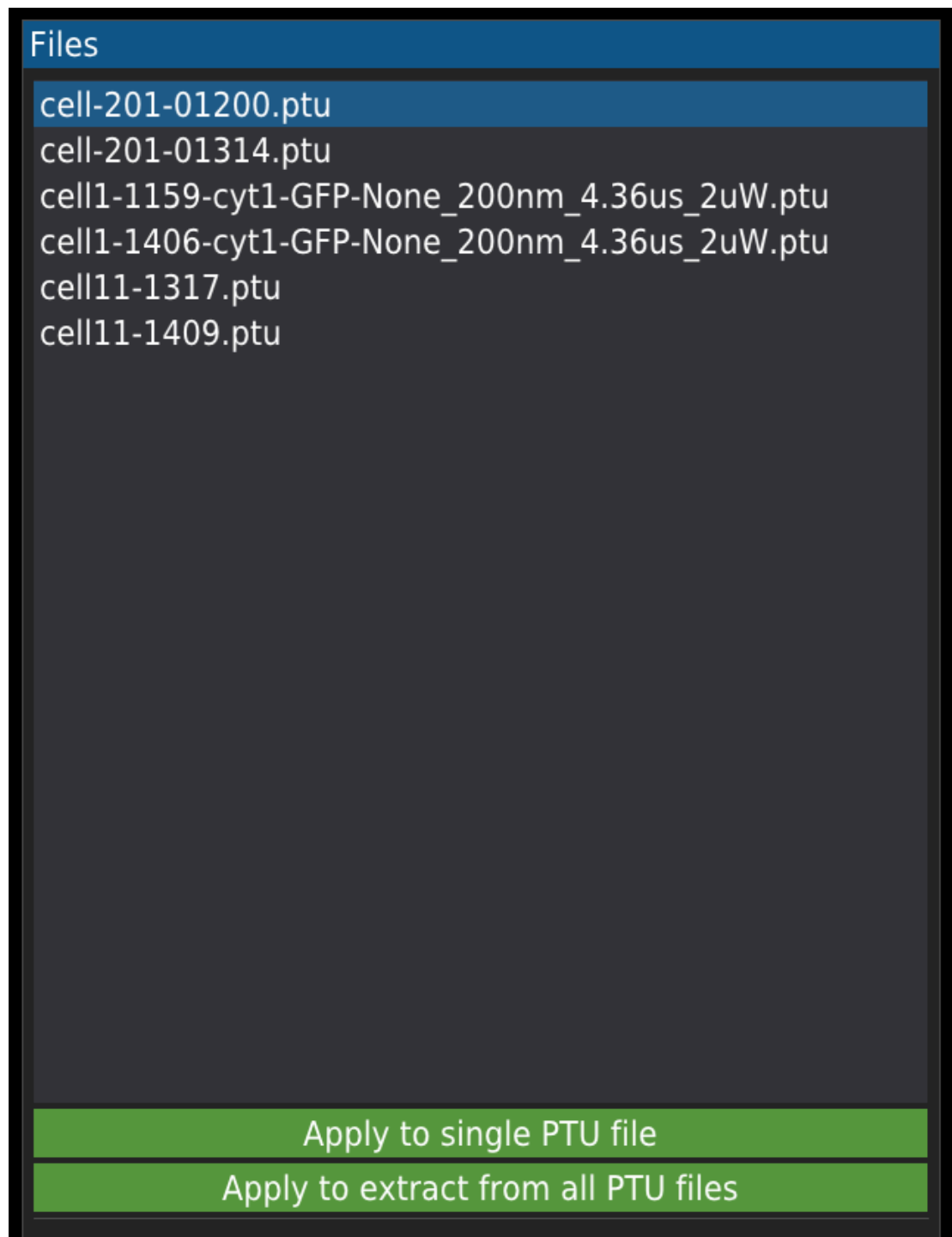
Only the signal corresponding to the lower plots will be used for analysis.

Important! Some older hardware based on NIKON A1 and LSM upgrade kits from PicoQuant may experience a hardware bug occurring during the crosstalk between the A1 controller and the SymPhoTime software for the .ptu files acquisition. The bug occurs as additional lines in the figure are collected by SymPhoTime, resulting in a seared-like image at the top of the picture. The panel displayed below, located at the top right corner of the main window, allows you to correct the additional lines.



Apply filtering and extract

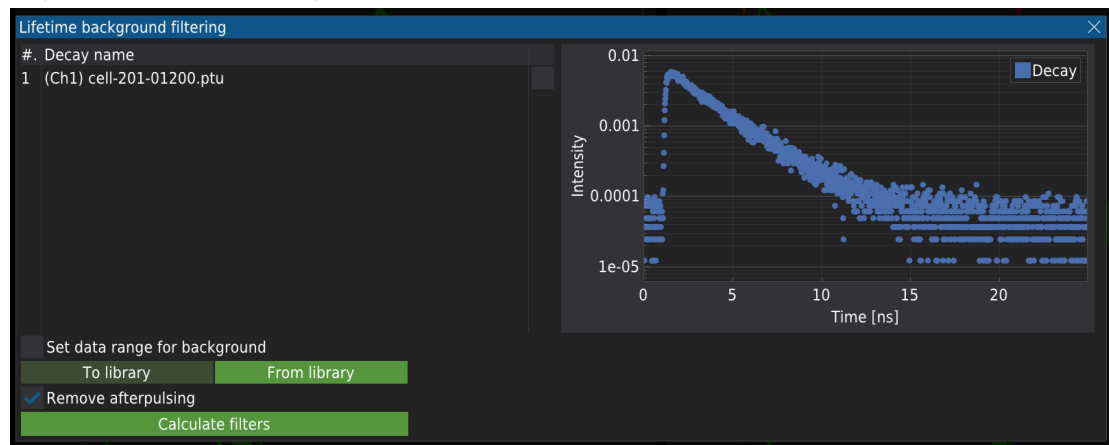
The panel displayed below shows the files located in the analyzed folder. Press "Apply to single PTU file" to analyze a single file. Pressing "Apply to extract from all PTU files" applies the selected analysis procedure (time gates or statistical filters) to all files from the list.



Statistical filters

Checking the "Use statistical filters" checkbox under each channel deactivates draglines and activates the "Calculate filters" button. This button opens the window (see below) where you can calculate statistical filters for background or fluorescence decay patterns

imported from the library.

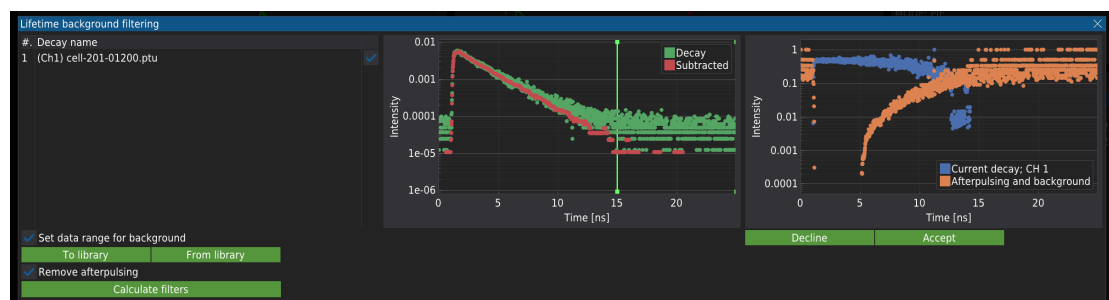


Removing background and afterpulsing

On the left of the above window is a list of decay patterns. Initially, only one pattern appears, corresponding to the pattern extracted from the image at a given channel; the one is displayed in the plot.

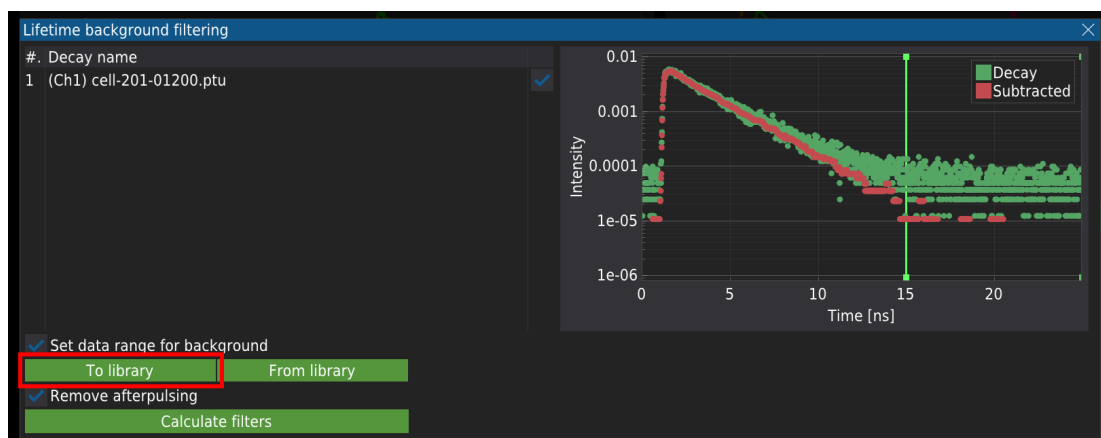
Selecting the "Set data range for background" checkbox activates draglines, allowing one to select the background region of the pattern, usually the plateau at the highest times of the decay pattern. Based on the selected region, the mean value of the background signal is calculated and subtracted from the original decay data. The subtracted data is further used for calculation of the statistical filter or stored in the library by pressing "To library"; [procedure will be described later](#).

Pressing the "Calculate filters" button will calculate the proper weights according to the procedure described in the literature.^{1,2} After that a new plot will appear. On the plot, one can see all the weights calculated for the curves in the left table, together with the background and afterpulsing signal. To accept the current set of weights, press "Accept"; to reject and recalculate, press "Decline".



Export the decay pattern to library

Instead of filter calculation, one can store the given decay pattern in the library for further use. For this purpose, instead of "Calculate filters," press "To library".

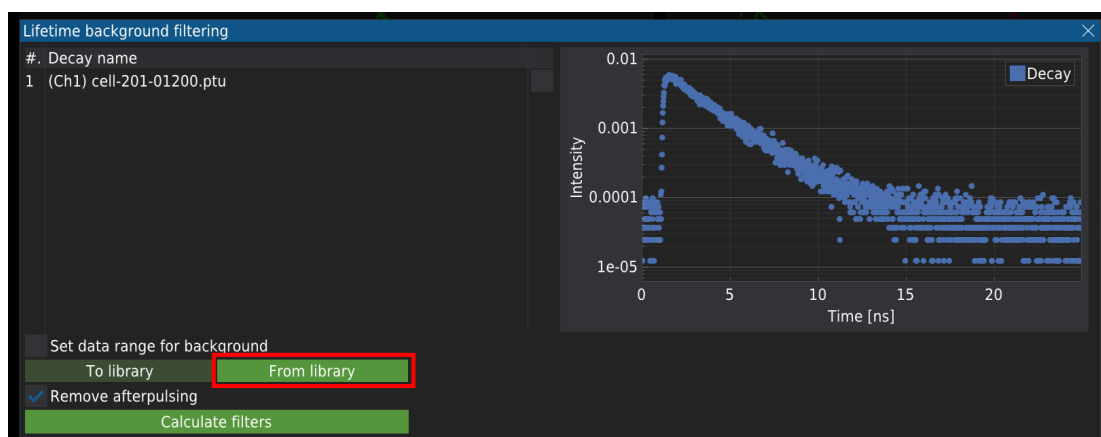


Below The "Calculate filters" button, you will see the new fields required to store the data.

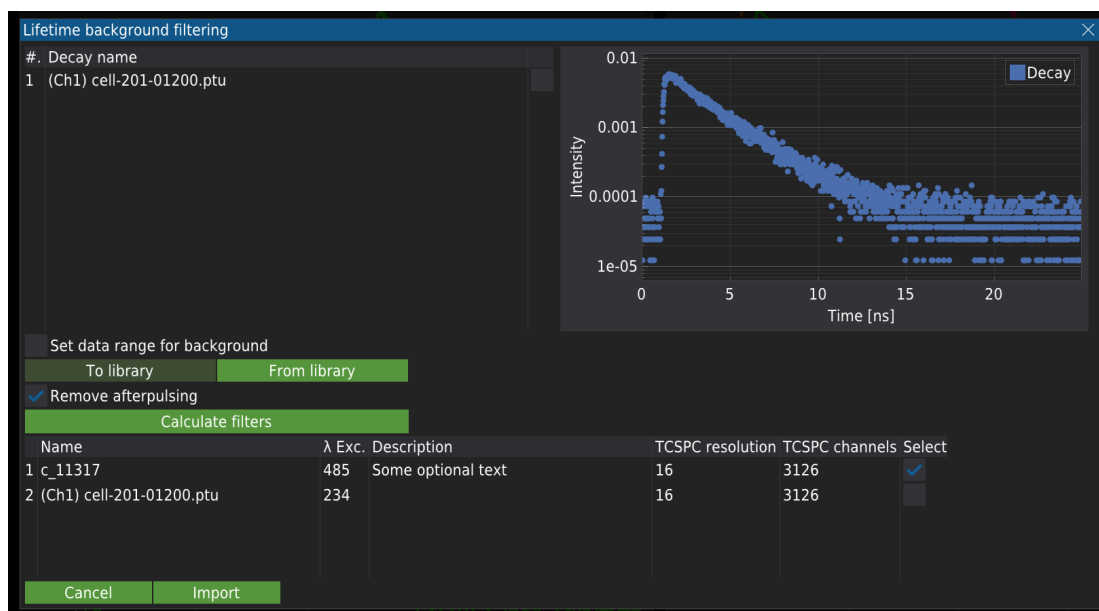
Although some fields are optional, the more details you provide, the easier you will find the proper decay. To store the data, press "Submit".

Import the decay pattern from library

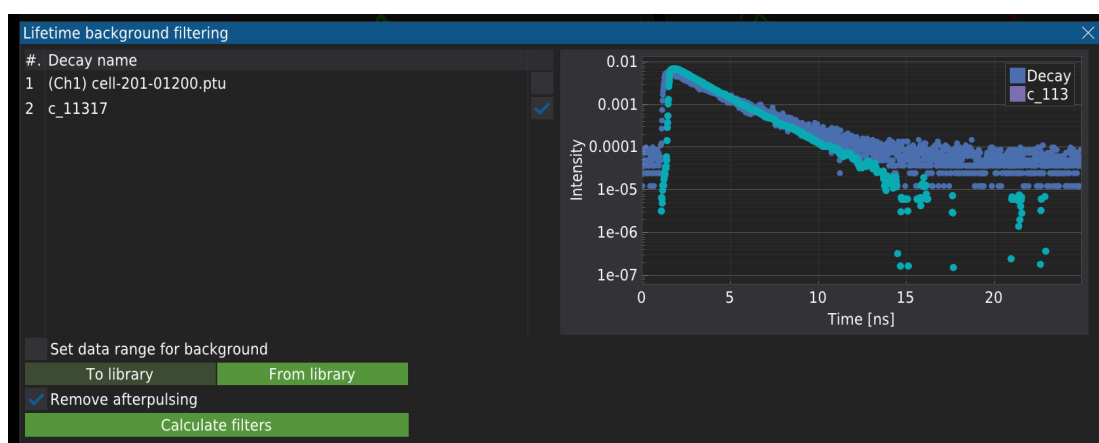
You can import stored decay data by pressing the "Form library" button.



Below the "Calculate filters" button, you will see the table containing all the stored decay patterns. Marking the dacay and pressing "Import" will load the dacay pattern.



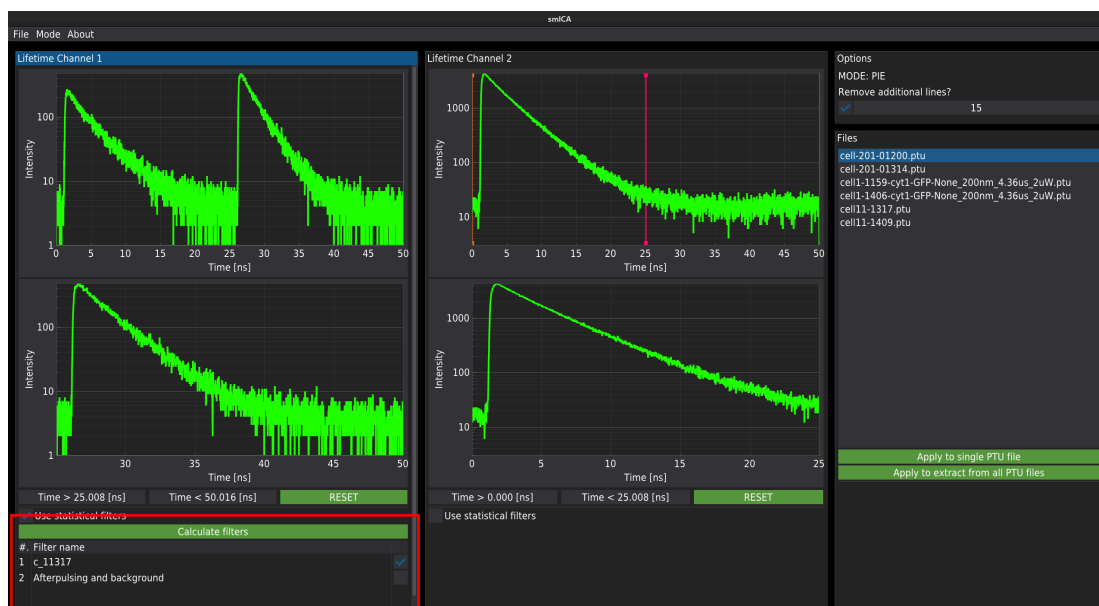
Note you can select many decay patterns, but they do not need to be physically justified.



You may also want to subtract the background (see [above](#)). Pressing "Calculate filters" will open the plot with calculated weights. Note the calculations can take some time. When you select many patterns, you should be sure which weight you would like to use for further analysis. Assuming that we are interested in detecting only the photons that originate from the imported decay here noted as *c_11317*, for further analysis, you will use the weight of the same name.

Filtering

Pressing the "Accept" button will close the window, and you can select the weights for analysis.



The same procedure can be repeated for Channel 2. Next, you can [apply](#) the filtering to the data.

OUTPUT

As an output, for each *.ptu* file, the script will generate several files required for further procedure steps. Assuming that the name of your file is *some-file.ptu*. The script will generate:

1. *some-file.pkl* - The binary file containing experimental data. The file is necessary to generate the concentration map.
2. Two *.png* files per channel.

The *.pkl* file is a pickle file containing the python dictionary. The structure of the file is described in the table below.

Keys	Value description
File info	The python dictionary containing: 'L_file' : name of the <i>.ptu_file</i> 'Pixels per line' : number of pixel in line in the image 'Number of lines' : number of lines in the image 'Pixels size' : size of the single pixel (nm) 'Number of frames' : number of frames acquired during the single measurement 'Pixel dwell' : The time in μs that is spent to illuminate a single pixel 'Lifetime resolution' : The temporal resolution of the fluorescence decay (ns)
export_df_1	The pandas DataFrame containing the number of photons registered in channel 1 for each pixel
export_df_2	The pandas DataFrame containing the number of photons registered in channel 2 for each pixel

Keys	Value description
taus_1	The pandas DataFrame containing the TCSPC histogram registered in channel 1. The time-gated signal
taus_2	The pandas DataFrame containing the TCSPC histogram registered in channel 2. The time-gated signal
fulltaus_1	The pandas DataFrame containing the full TCSPC histogram registered in channel 1.
fulltaus_2	The pandas DataFrame containing the full TCSPC histogram registered in channel 2.
lifetimes_1	Numpy array containing the fluorescence lifetime for each pixel registered in channel 1. Signal after time-gating/filtration.
lifetimes_2	Numpy array containing the fluorescence lifetime for each pixel registered in channel 2. Signal after time-gating/filtration.
intensity_1	Numpy array containing the number of photons for each pixel registered in channel 1. Signal after time-gating/filtration.
intensity_2	Numpy array containing the number of photons for each pixel registered in channel 2. Signal after time-gating/filtration.

The minimal python script to get the pickled data is below.

```
import os
import pickle
import pandas as pd
import numpy as np
path = '<path-to-file>/<file-name>.pkl'

with open(path, 'rb') as file:
   .pkl = pickle.load(file)

file_info = .pkl['File info']
print(file_info)

DF_1 = .pkl['export_df_1']
DF_2 = .pkl['export_df_2']
print(DF_1)
print(DF_2)

DF_taus_1 = .pkl['taus_1']
DF_taus_2 = .pkl['taus_2']

print(DF_taus_1)
print(DF_taus_2)

DF_ftaus_1 = .pkl['fulltaus_1']
DF_ftaus_2 = .pkl['fulltaus_2']

print(DF_ftaus_1)
print(DF_ftaus_2)

lifetimes_1 = .pkl['lifetimes_1']
lifetimes_2 = .pkl['lifetimes_2']
```

```

print(lifetimes_1)
print(lifetimes_2)

intensity_1 = pkl['intensity_1']
intensity_2 = pkl['intensity_2']

print(intensity_1)
print(intensity_2)

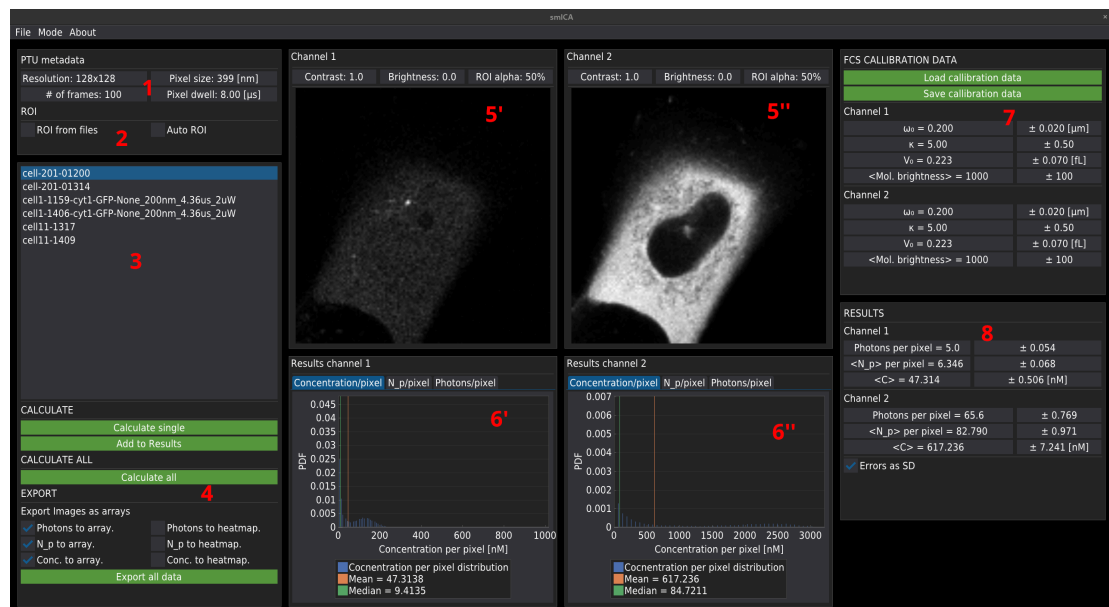
```

Phot 2 Conc

This mode allow to map the concentration of fluorescent molecules using single-molecule FLIM imaging data. The script requires the *.pkl* files created in the [EXTRACT from PTU and FILTER](#) mode. **Important! This script does not read the *.ptu* files directly. It is necessary to use the [EXTRACT from PTU and FILTER](#) mode first!**

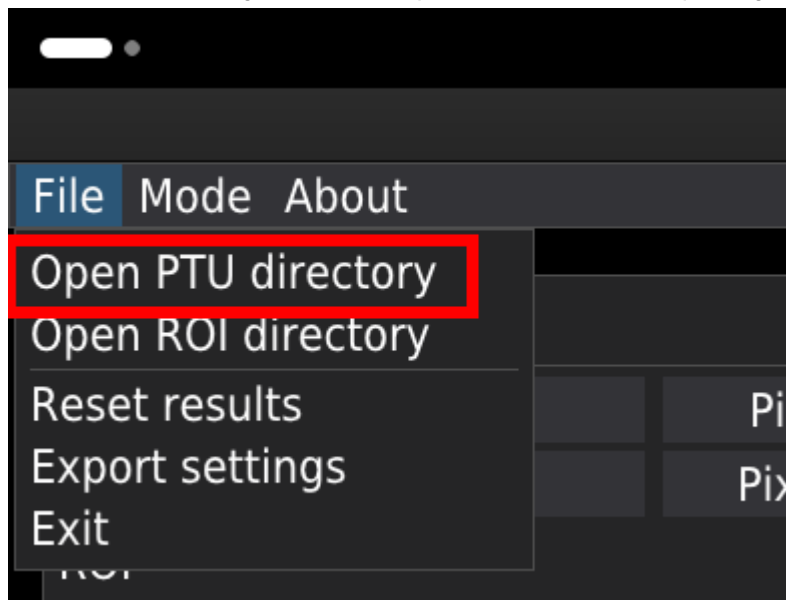
The GUI is divided into several sections.

1. PTU metadata containing the information about image acquisition.
2. ROI containing the selction panel for coosing the metod for the region of interest. It also allow to controll the automatic ROI [see below](#).
3. The list of files for analysis that are located in the PTU folder.
4. Buttons starting the claculations and export oppitons.
5. Analysed image - 5' for channel 1 and 5'' for channel 2.
6. Histograms of the conentration numberof molecules or mubmber of photons per pixel - 6' for channel 1 and 6'' for channel 2.
7. The FCS callibration data
8. Mean results.



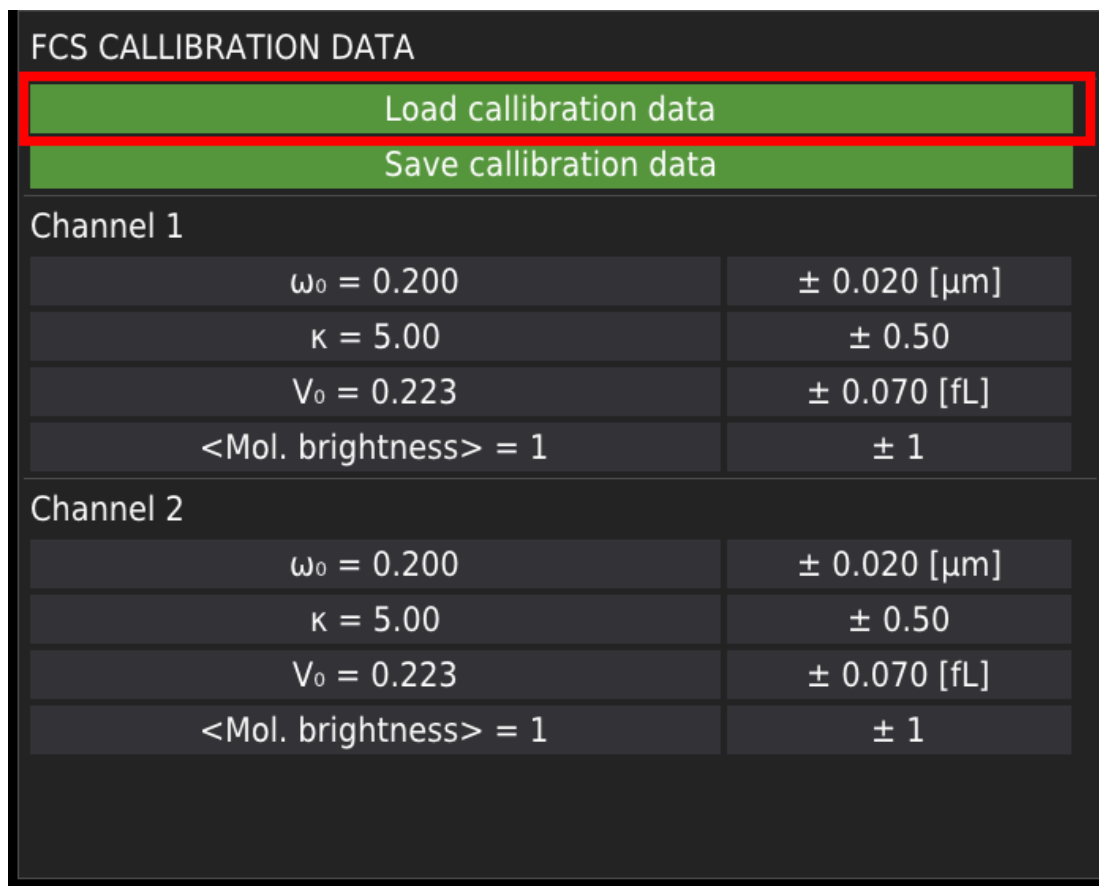
Loading files and callibration data

The folder containing files for analysis can be selected by using File menu.



Note. The folder should contain only *.ptu* files and the corresponding *.pkl* files.

After loading the list of files it is recommended to load or input the FCS calibration data and molecular brightness of the fluorophore required for proper calculation of the number of molecules and concentration.



or manually.

FCS CALIBRATION DATA

Load calibration data

Save calibration data

Channel 1

$\omega_0 = 0.200$	$\pm 0.020 \text{ } [\mu\text{m}]$
$\kappa = 5.00$	± 0.50
$V_0 = 0.223$	$\pm 0.070 \text{ [fL]}$
$\langle \text{Mol. brightness} \rangle = 1$	± 1

Channel 2

$\omega_0 = 0.200$	$\pm 0.020 \text{ } [\mu\text{m}]$
$\kappa = 5.00$	± 0.50
$V_0 = 0.223$	$\pm 0.070 \text{ [fL]}$
$\langle \text{Mol. brightness} \rangle = 1$	± 1

Note. When you load the file it needs to have a proper structure. For more details check the Callibration_exmample files (.json) in the smaples folder. The calibration values provided manually can be saved by pressing the save button

FCS CALIBRATION DATA

Load calibration data

Save calibration data

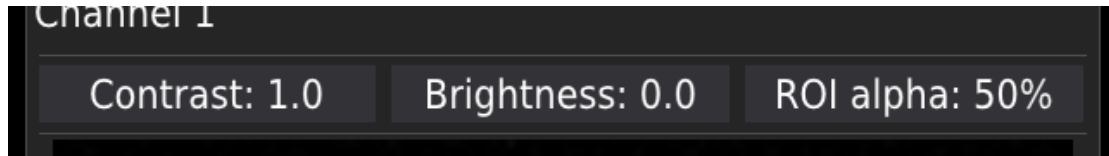
Channel 1

$\omega_0 = 0.200$	$\pm 0.020 \text{ } [\mu\text{m}]$
$\kappa = 5.00$	± 0.50
$V_0 = 0.223$	$\pm 0.070 \text{ [fL]}$
$\langle \text{Mol. brightness} \rangle = 1$	± 1

Channel 2

$\omega_0 = 0.200$	$\pm 0.020 \text{ } [\mu\text{m}]$
$\kappa = 5.00$	± 0.50
$V_0 = 0.223$	$\pm 0.070 \text{ [fL]}$
$\langle \text{Mol. brightness} \rangle = 1$	± 1

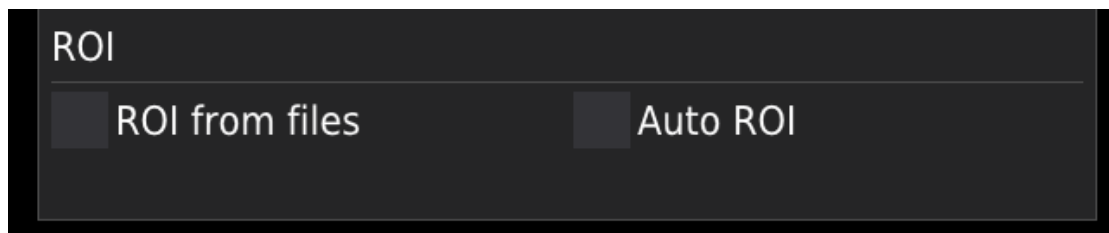
The analysed image is displayed in windows corresponding to channel 1, 2 or both. Above the images the user is able to adjust the visibility of the image by changing contrast and brightness or the overlaying ROI's alpha channel; [see ROI section](#).



The **adjustment** is performed only on the displayed image (channel 1 or channel 2) and **does not influence the results**.

Selecting the ROI

The region of interest is a part of the image that will be considered during the image analysis. It is marked as the red area on the image. The software has two options for ROI definition: "ROI from files" or "Auto ROI".



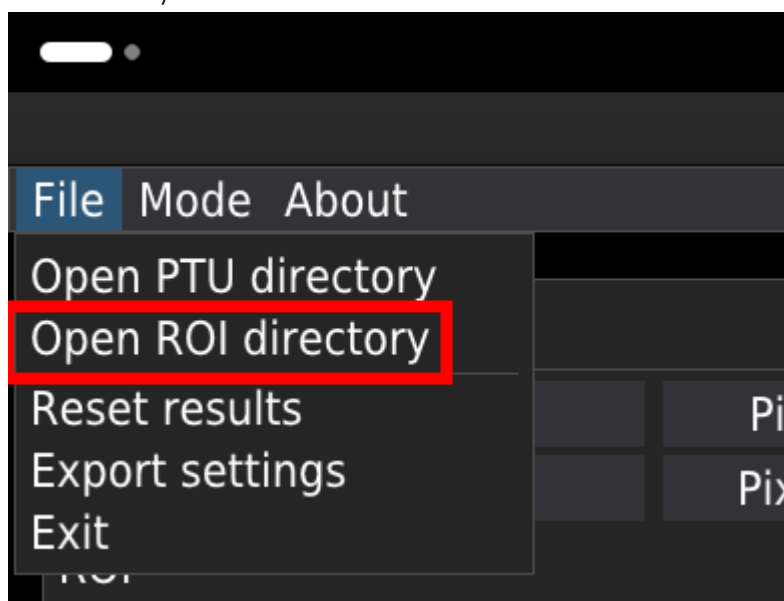
The "ROI from files" option requires externally created roi files. The examples of externally created ROI files (using ImageJ) are given in the samples/ROI folder. The files should be rewritten with the *REWRITE_ROI.py* [script](#) before import. Note that (i) each channel's ROI file is created separately. (ii) the filename of the ROI file should have the same structure as the PTU file with an additional string at the end of the filename, according to the example given below.

file_name.ptu

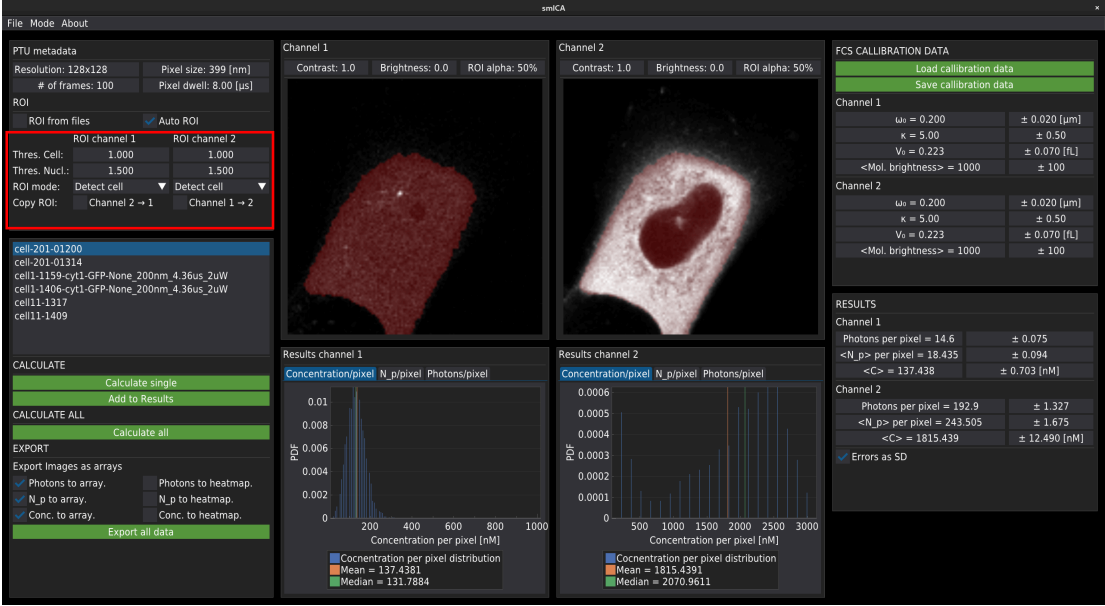
file_name_roi_ch_1.dat - (ROI file for channel 1)

file_name_roi_ch_2.dat - (ROI file for channel 2)

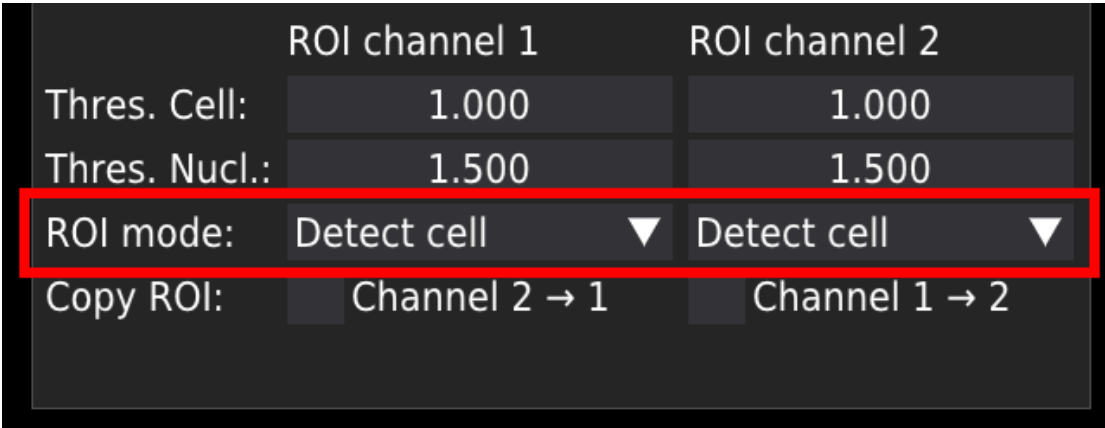
The ROI files will automatically load the given *.ptu* file. To import the ROI files, the user must specify the folder where the files are located by selecting the path using the "Open ROI directory" from the File menu.



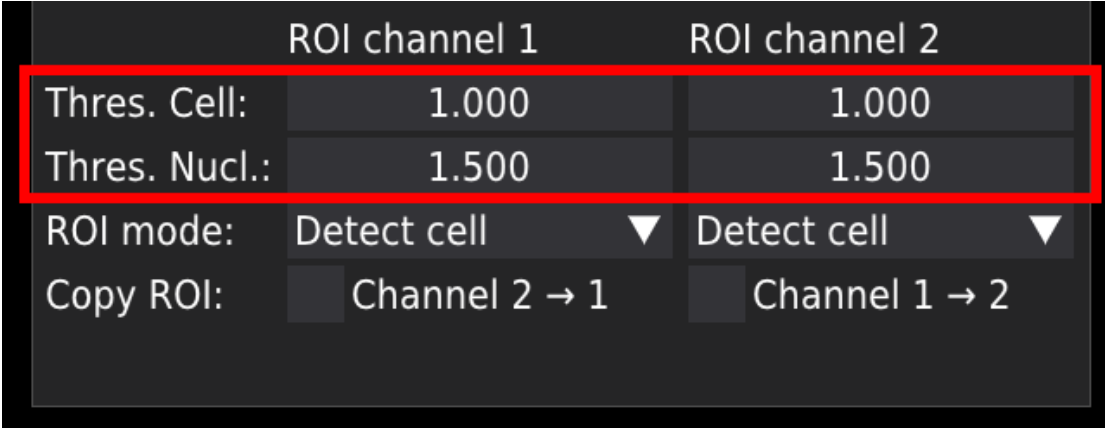
The second option for selecting the region of interest is the automatic mode. Selecting the Auto ROI checkbox displays the control panel for the ROI in both channels and the ROI on the image as a red field.



At this moment, the automatic ROI option offers two modes of ROI detection: "Detect cell" and "Subtract nucleus." The modes can be selected for each channel independently by selecting from the dropdown menu (see below).



Both modes use the OTSU thresholding method. The user can independently shift the original OTSU thresholding level by changing each channel's "Thres. cell" or "Thres. Nucl." sliders.



Default value for "Thres. cell" sliders is 1, while for "Thres. Nucl." is 1.5. A value equal to 1 means no change concerning the original OTSU thresholding level. The user can change

the thresholding levels to find the desired ROI. This procedure need to be done for each file separately. To subtract the nucleus from the cell ROI, it may be necessary to increase the value of the "Thres. Nucl." to around 2.

The automatically found ROI can be copied between channels by selecting "Channel 1 -> 2" or "Channel 2->1" checkboxes.

	ROI channel 1	ROI channel 2
Thres. Cell:	1.000	1.000
Thres. Nucl.:	1.500	1.500
ROI mode:	Detect cell ▼	Detect cell ▼
Copy ROI:	Channel 2 → 1	Channel 1 → 2

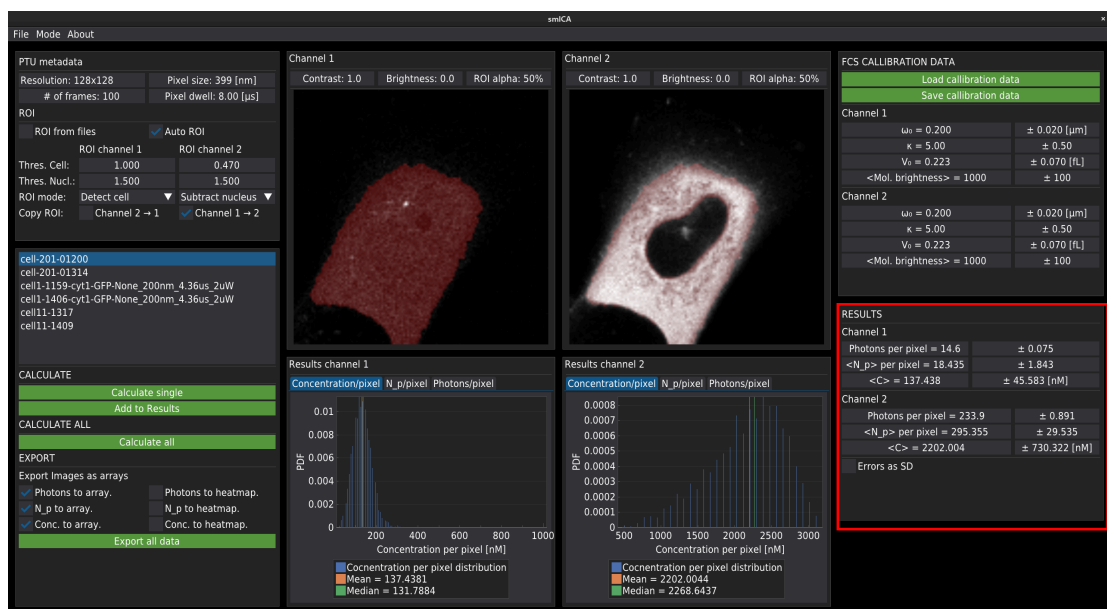
The copying is performed only for the "detect cell" mode. Note that when the checkbox is selected, the thresholding levels can be regulated only in the original channel. For example, selecting "Channel 1->2" will copy the ROI from channel 1 to channel 2, but the "Thres. cell" values can be controlled only for channel 1. The copy ROI option will not affect the threshold for nucleus subtraction; namely, the nucleus will be detected and subtracted from the ROI copied from channel 1.

Calculate results

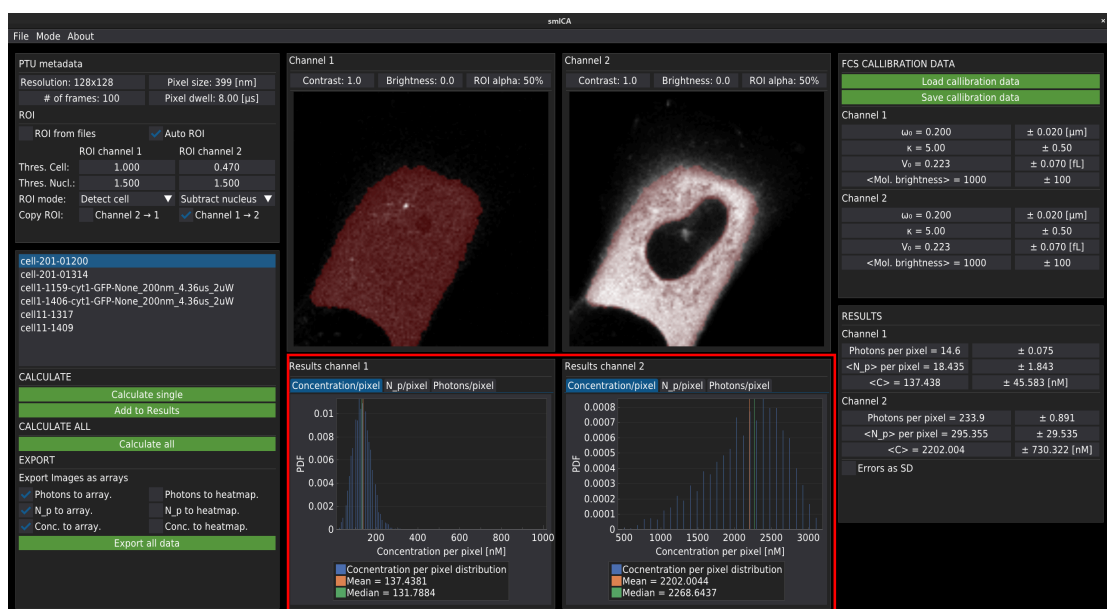
To correctly calculate concentration maps, the user must provide or import the calibration data obtained from FCS measurements, including the width of the focal volume ω_0 , the structure parameter κ , and the mean molecular brightness; [see above](#). Pressing "Calculate single" will display the results for a single file. Press the "Add to results" button to store the results and remember the Auto ROI settings for a given file. The calculate all button will automatically calculate and store the data for all files, one by one.

CALCULATE	
Calculate single	
Add to Results	
CALCULATE ALL	
Calculate all	

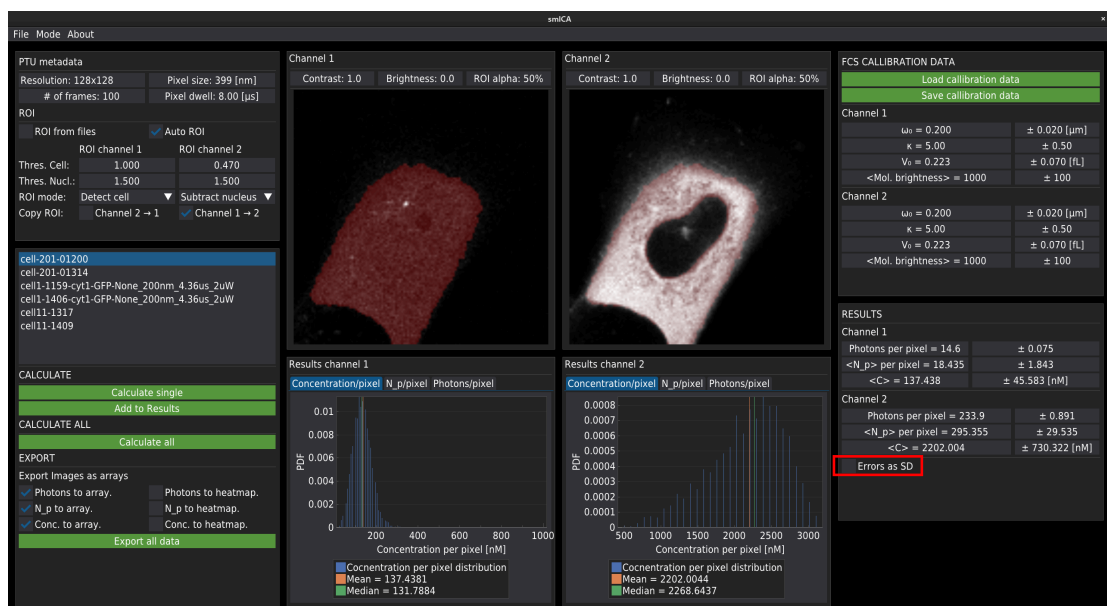
The result will be displayed in these tables:



and presented in the form of concentration, N_p , or photon distribution:



The errors displayed in the RESULTS window can be calculated in two ways. Errors as SD switch changes between them:

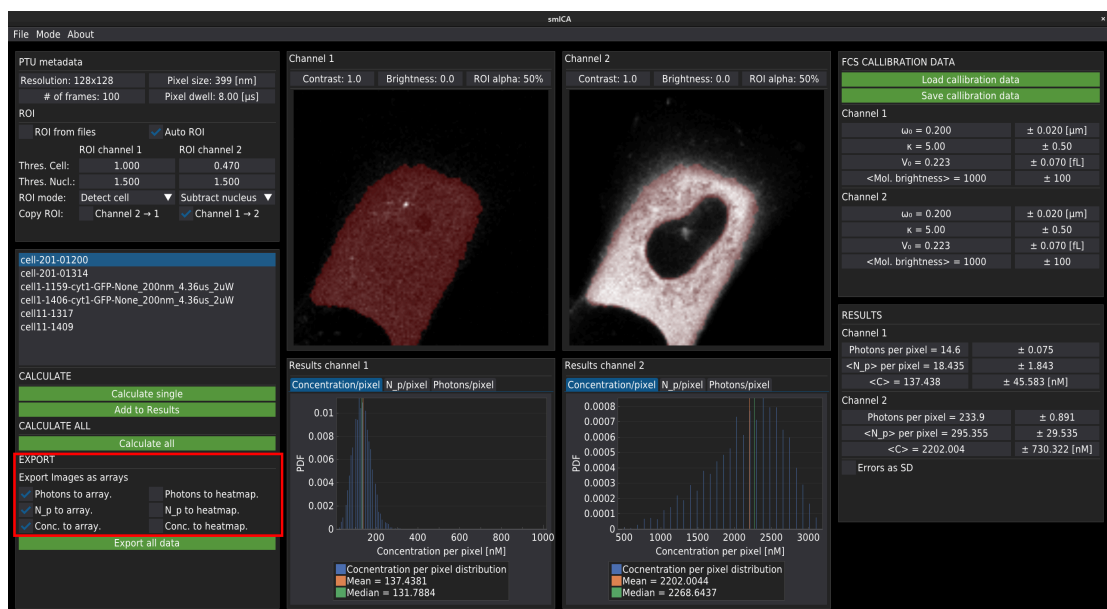


In the first way (unmarked checkbox), the error is returned as the mean over the maximal errors calculated for each pixel. Those errors include errors for V_0 and the molecular brightness.

The second error calculation method (marked checkbox) returns the standard deviation SD , calculated from all pixels, and divided by the square root of the number of all pixels.

Export results

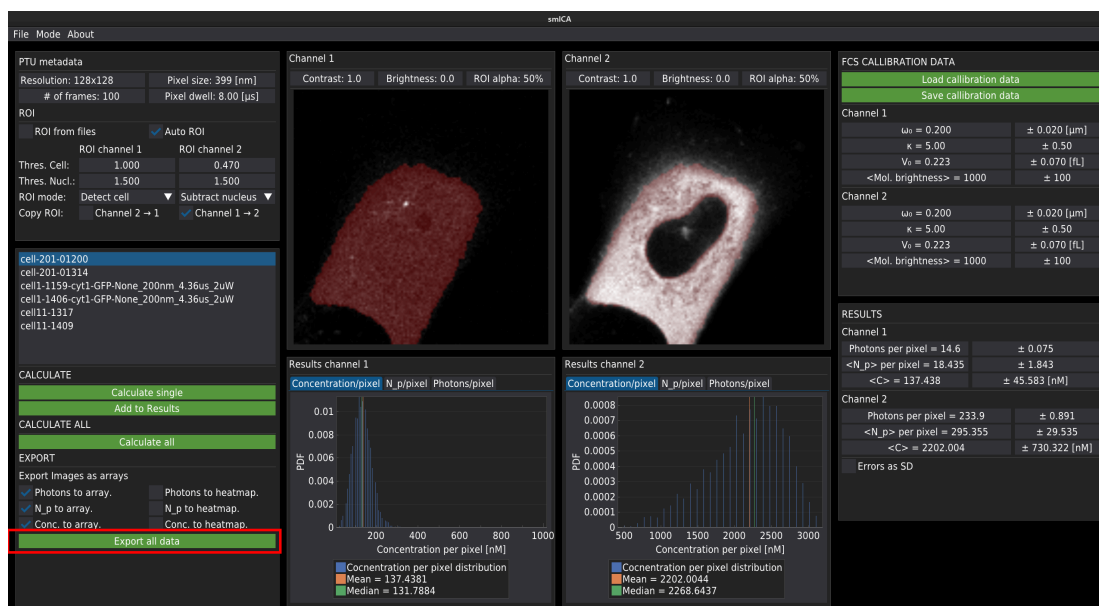
During calculation, the data will be automatically exported according to rules marked in the export panel.



"to array" means that it will create the file containing the array (size of the image) containing a number of photons, molecules, or concentration in a given pixel.

"to heatmap" means that the heatmap will be exported as the *.png* file.

To export the averaged data as a table, press the "Export all data" button.



References

- [1] Enderlein, Jörg, and Ingo Gregor. "Using fluorescence lifetime for discriminating detector afterpulsing in fluorescence-correlation spectroscopy." Review of scientific instruments 76.3 (2005).
- [2] Kapusta, Peter, et al. "Fluorescence lifetime correlation spectroscopy." Journal of Fluorescence 17 (2007): 43-48.

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